

Lauriane Mbagdje Dorenan

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PROFILE

Artificial Intelligence Engineer with hands-on experience in machine learning, deep learning, computer vision, and generative AI systems. Skilled in building, fine-tuning, and deploying AI models for healthcare and real-world applications. Strong expertise in Python-based ML pipelines, dataset curation, and multimodal AI systems.

TECHNICAL SKILLS

Programming: Python, R
ML/DL: Scikit-learn, TensorFlow, Keras, PyTorch, Transformers, ViT, LoRA, Quantization
Computer Vision: CNNs, Image Classification, Data Augmentation, OCR Pipelines
NLP & GenAI: LLaMA fine-tuning, Whisper (ASR), NLLB-200, Prompt Engineering
Tools: Pandas, NumPy, Git, Docker, Streamlit, Hugging Face
Languages: French (C1), English (B2)

PROFESSIONAL EXPERIENCE

AfriAI Solutions | [Link](#) Dakar, Senegal
Co-founder & Data Scientist (Part-time) Apr 2025 – Present

- Designed and implemented AI prototypes including multimodal health chatbots and computer vision systems.
- Built end-to-end ML pipelines: data preprocessing, model training, evaluation, and deployment.

eath Startup | [Link](#) Senegal
AI Engineer – Food Safety Application 2024 – Present

- Contributed to ingredient detection pipelines using computer vision techniques.
- Curated and annotated datasets for allergen and ingredient classification tasks.

SSM Systèmes Médicaux SA | [Link](#) Dakar, Senegal
CMMS Manager Apr 2025 – Present
Biomedical Equipment Technician Intern Aug 2024 – Mar 2025

- Analyzed structured maintenance datasets to support predictive and preventive strategies.
- Improved operational reporting and data-driven decision-making processes.

PROJECTS

Skin Disease Classification (Vision Transformer – ViT) | [Link](#)

- Fine-tuned ViT-base for 22-class dermatological classification (98.49% accuracy).

MedBot – Multimodal Health Chatbot | [Link](#)

- Fine-tuned LLaMA-3.2-3B-Instruct using LoRA and 4-bit quantization.
- Integrated Whisper ASR and multilingual translation (NLLB-200).

EduConnect – AI Academic Orientation Platform | [Link](#)

- Contributed to AI-driven academic recommendation workflows and chatbot logic.

COVID-19 X-ray Classification App | [Link](#)

- Developed CNN (Keras) achieving 88% accuracy; deployed via Streamlit.

EDUCATION

Master in Artificial Intelligence — Dakar Institute of Technology 2023–2025
Bachelor in Biomedical Engineering — ESP-UCAD 2019–2022

CERTIFICATIONS

Artificial Intelligence Fundamentals with Capstone Project (IBM) 2025
AI Programming with Python (Udacity) 2024
Azure AI Fundamentals (Microsoft) 2024